
Introduction to Docker

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Docker images :

```
FROM ubuntu  
RUN apt-get update  
CMD [ "echo" , "hello from container" ]
```

Docker image from build to runtime :

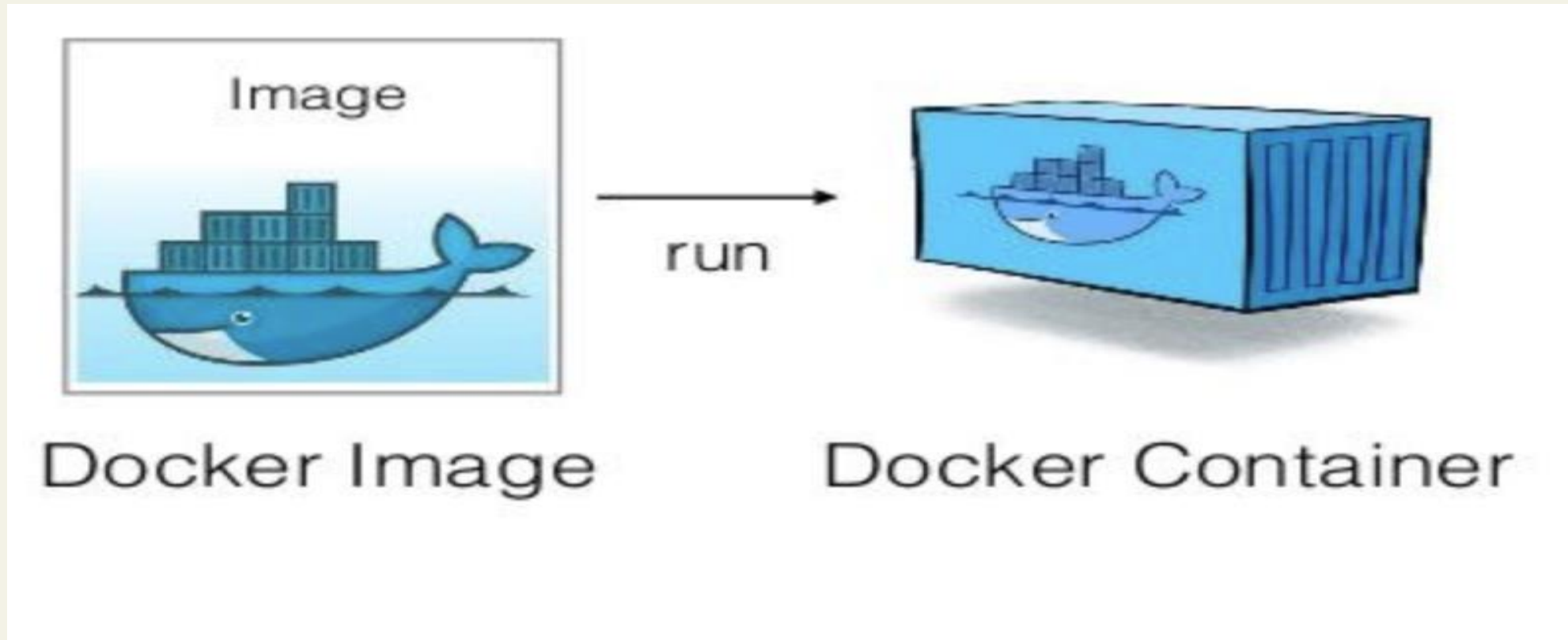
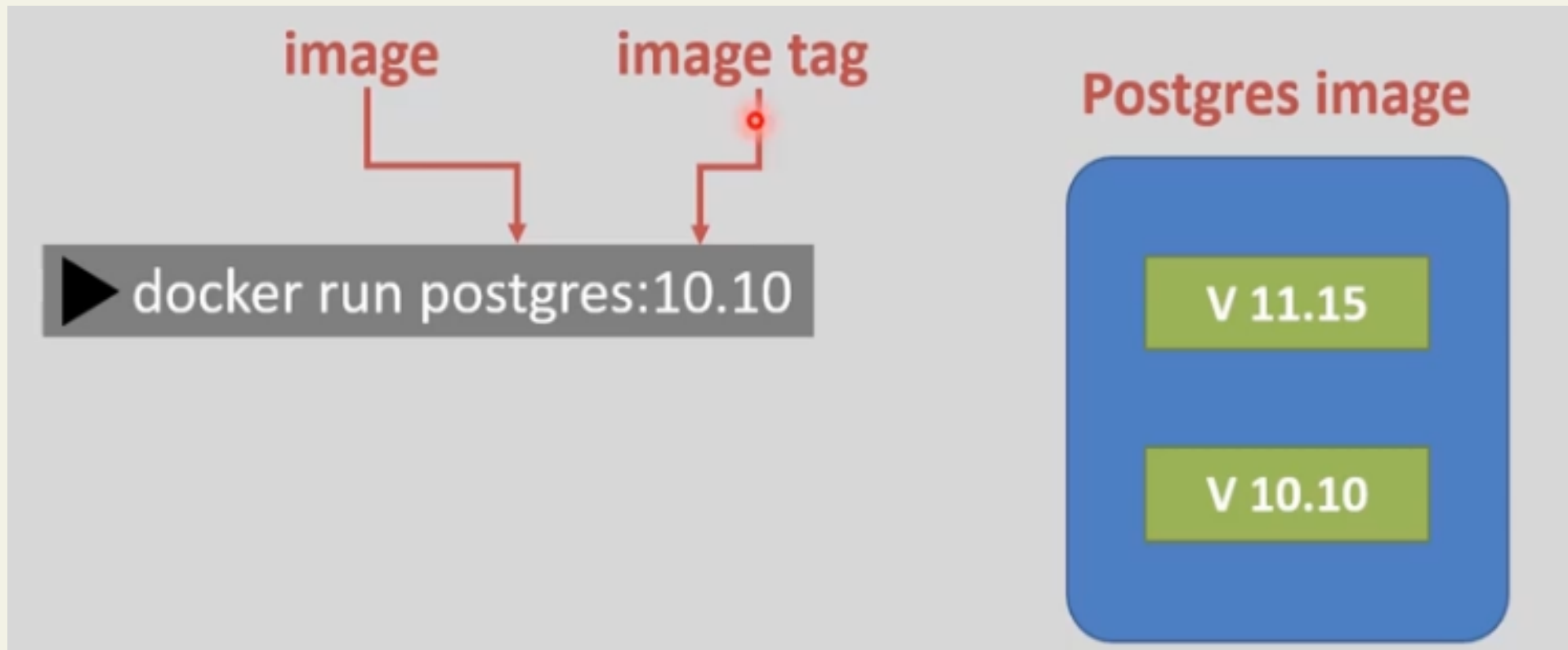


Image tag:



Lab6:



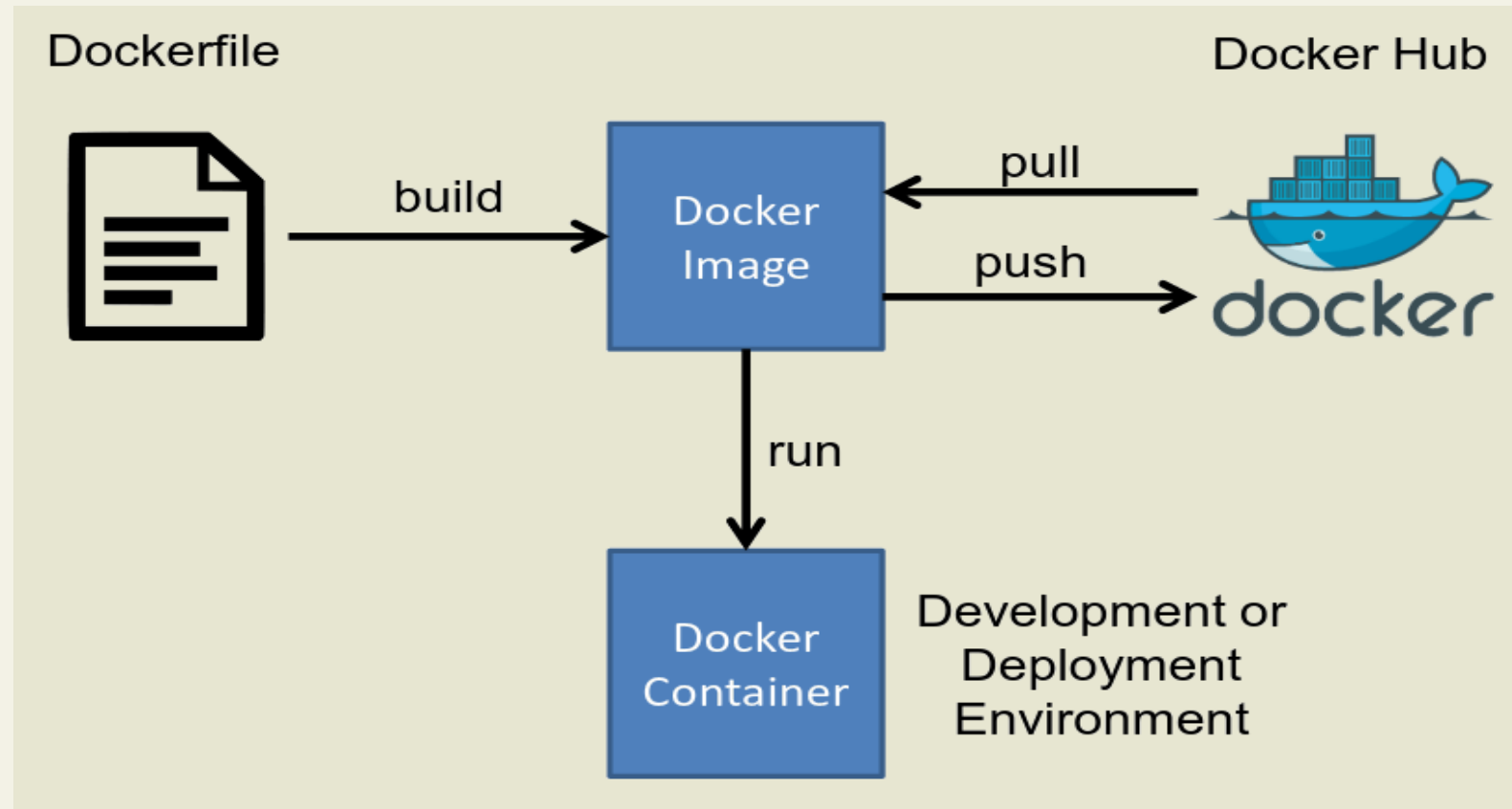
Docker images(cont.):

Docker run postgres:latest

Docker run postgres

docker history <image-name>

Docker registry:



Docker registry commands:

```
docker login
```

```
docker tag local-image:tagname [username-account]/repo-name:tagname
```

```
docker push [username-account]/new-repo:tagname
```

Lab7:



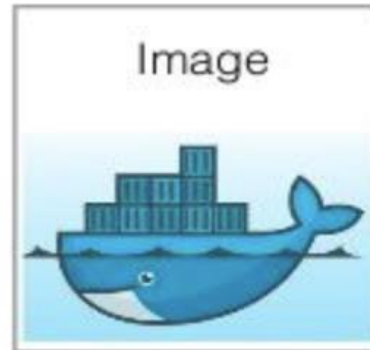
Docker file:

```
docker build -t <image-name> .
```

```
FROM ubuntu:14.04
MAINTAINER John Doe <john.doe@example.com>
RUN apt-get update && apt-get install -y python
RUN python --help
```

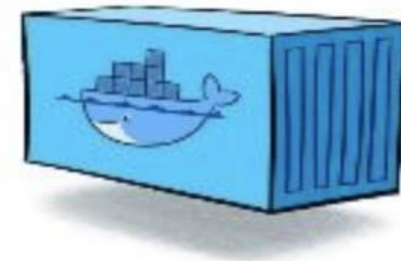
Dockerfile

build



Docker Image

run



Docker Container

Docker file structure:

1- FROM <image>:[tag]

2- Run <command>

3- CMD ["executable", "Raram1" , "Raram2"]

```
FROM ubuntu  
RUN apt-get update  
CMD [ "echo" , "hello from container" ]
```

Lab8:



Docker file (cont.):

1- ENTRYPOINT ["executable"]

2-MAINTAINER <name>

3- EXPOSE <port>

```
FROM ubuntu  
RUN apt-get update  
MAINTAINER ranaahmed  
ENTRYPOINT [ "echo " ]
```

Docker file (cont.):

1- COPY <source> <destination>

2- WORKDIR </path >

3- ENV <key>= <value>

Lab9:



Data management in Docker:

There are two ways to persist data in docker :

1- docker volumes

2- bind mounting

1- docker volumes:

- ☐ Volumes are stored in a part of the host filesystem which is managed by docker filesystem which is managed by docker (/var/lib/docker/volumes/ on linux).
- ☐ Volumes are the best way to persist data in Docker.
- ☐ A given volume can be mounted into multiple containers

Docker volumes commands:

```
docker volume create <volume-name>
```

```
docker volume ls
```

```
docker volume rm <volume-name>
```

```
docker run -it -d -v [volume-name]:[container dir] <image-name>
```

```
docker run -it -d -v [volume-name]:[container dir]:ro <image-name>
```

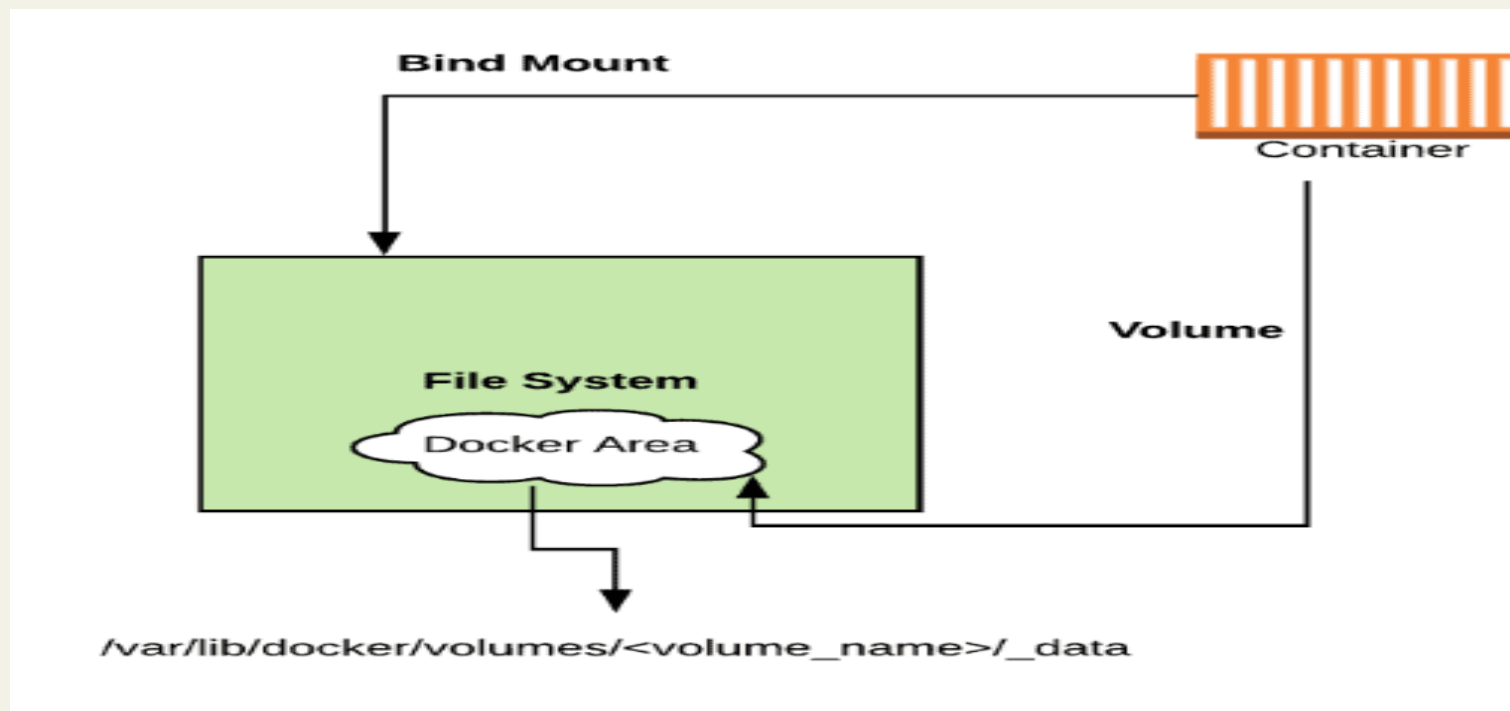
lab10:



Lab

2-bind mounting:

- ❑ Bind mounts may be stored anywhere on the host system.
- ❑ `docker run -it -d -v [host directory]:[container-dir] <image-name>`



Lab11:



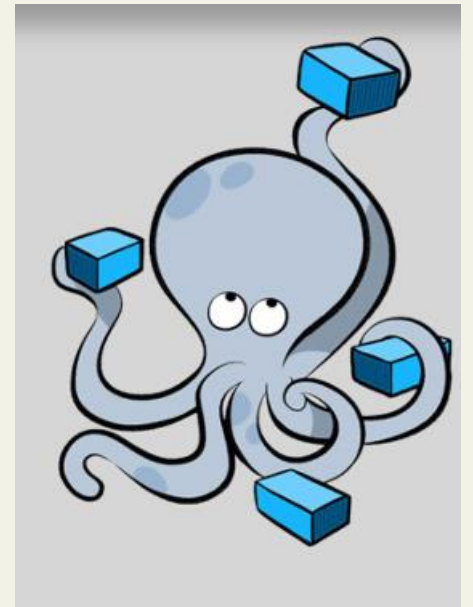
Docker compose:

Install docker compose :

1- `sudo curl -L "https://github.com/docker/compose/releases/download/1.29.2/docker-compose-$(uname -s)-$(uname -m)" -o /usr/local/bin/docker-compose`

2- `sudo chmod +x /usr/local/bin/docker-compose`

3- `docker-compose --version`



Docker compose file structure:

Version from that : <https://docs.docker.com/compose/compose-file/compose-versioning/>

```
version: '3'
services:
  database:
    image: mysql
    container_name: test
    volumes:
      - hostpath:containerpath
      - volumename:containerpath
  frontend:
    build: .
    ports:
      - "3000:5000"
volumes:
  volumeName: {}
```

Docker compose commands:

```
docker compose up
```

```
Docker compose -f <file-name> up
```

```
Docker compose up -d
```

Docker compose commands:

Docker compose down

Docker compose down - -volumes

docker compose ps

Lab12:



Lab



The End